In [ ]:
```
EXPERIMENT 6 – DATA VISUALIZATION WITH SEABORN
PIN NO: 241UDS0109

AIM:-
To visualize and analyze a dataset using Seaborn plots including scatter plot, KDE plot, histogram, count plot,
bar plot, boxplot, strip plot, joint plot, and heatmap.
```

In [ ]:
```
## Step 1 – Import the Required Libraries & Dataset
```

In [3]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [4]:
```
df_241UDS0109 = pd.read_csv("Experiment_6_241UDS0109_City_Bike_Rentals.csv")
df_241UDS0109.head()
```

Out[4]:

	city	date	rentals	state
0	Visakhapatnam	2022-01-12	820	Andhra Pradesh
1	Visakhapatnam	2022-03-18	857	Andhra Pradesh
2	Visakhapatnam	2022-05-21	894	Andhra Pradesh
3	Visakhapatnam	2022-07-09	931	Andhra Pradesh
4	Visakhapatnam	2023-02-14	940	Andhra Pradesh

In [ ]:
```
## Step 2 – Info of the Dataset
```

In [5]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 80 entries, 0 to 79
Data columns (total 4 columns):
 #   Column   Non-Null Count  Dtype 
---  ------   --------------  ----- 
 0   city     80 non-null     object
 1   date     80 non-null     object
 2   rentals  80 non-null     int64 
 3   state    80 non-null     object
dtypes: int64(1), object(3)
memory usage: 2.6+ KB
```

In [ ]:
```
## Step 3 – State-wise Data Points
```

In [6]:
```
df_241UDS0109.value_counts("state")
```

Out[6]:
```
state
Andhra Pradesh    16
Karnataka         16
Maharashtra       16
Rajasthan         16
Tamil Nadu        16
Name: count, dtype: int64
```

In [ ]:
```
## Step 4 – Convert Date Column to Datetime
```

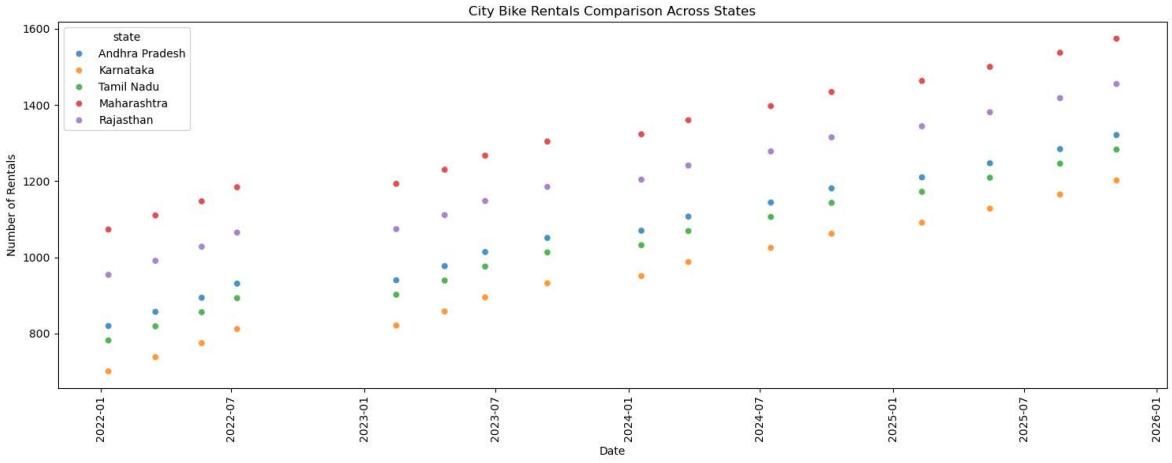
In [7]:
```
df_241UDS0109["date"] = pd.to_datetime(df_241UDS0109["date"])
df_241UDS0109["year_241UDS0109"] = df_241UDS0109["date"].dt.year
df_241UDS0109.head()
```

Out[7]:

	city	date	rentals	state	year_241UDS0109
0	Visakhapatnam	2022-01-12	820	Andhra Pradesh	2022
1	Visakhapatnam	2022-03-18	857	Andhra Pradesh	2022
2	Visakhapatnam	2022-05-21	894	Andhra Pradesh	2022
3	Visakhapatnam	2022-07-09	931	Andhra Pradesh	2022
4	Visakhapatnam	2023-02-14	940	Andhra Pradesh	2023

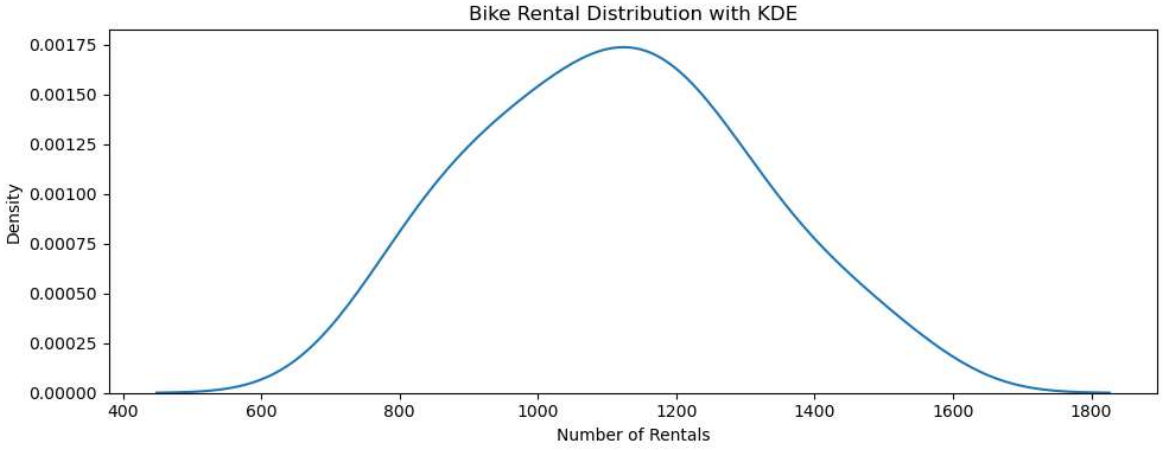
In [ ]:
```
## Step 5 – Visualize Data Using Various Plots
### Scatter Plot – Bike Rentals in Different States Over Time
```

In [8]:
```
plt.figure(figsize=(15, 6))
sns.scatterplot(
    x="date",
    y="rentals",
    hue="state",
    data=df_241UDS0109,
    alpha=0.8
)
plt.title("City Bike Rentals Comparison Across States")
plt.xticks(rotation=90)
plt.xlabel("Date")
plt.ylabel("Number of Rentals")
plt.tight_layout()
plt.show()
```



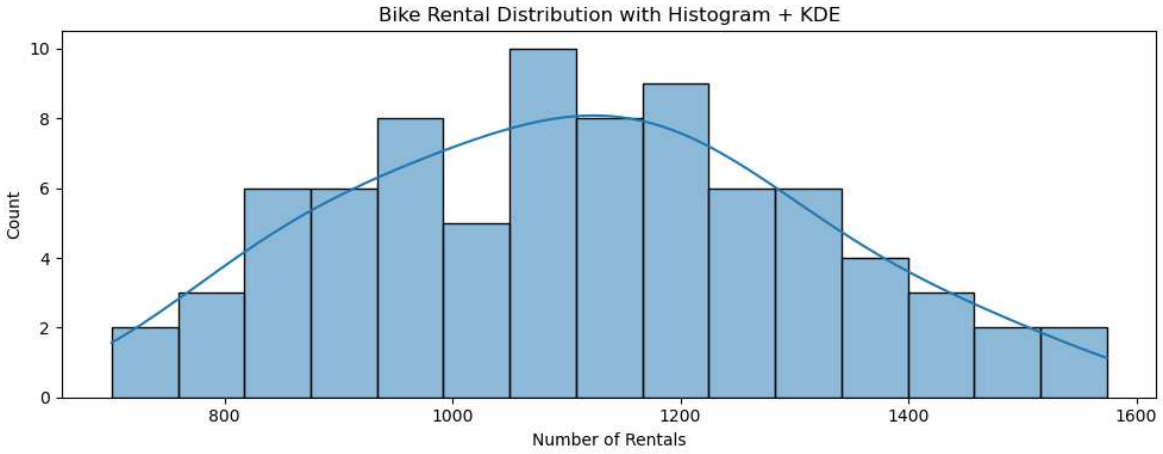
In [ ]:
```
## Step 6 – Histograms & KDE
### KDE Plot – Rental Distribution with KDE
```

In [9]:
```
plt.figure(figsize=(10, 4))
sns.kdeplot(data=df_241UDS0109, x="rentals")
plt.title("Bike Rental Distribution with KDE")
plt.xlabel("Number of Rentals")
plt.ylabel("Density")
plt.tight_layout()
plt.show()
```



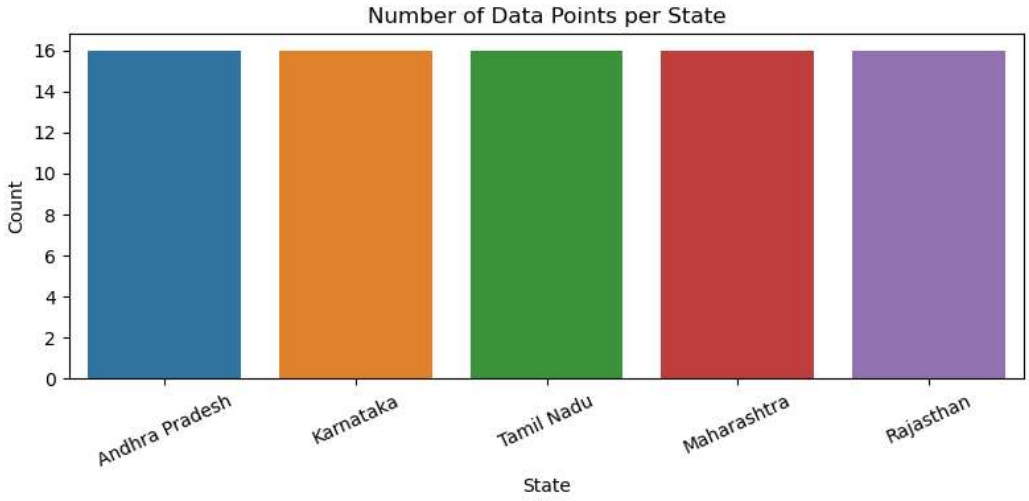
In [ ]:
```
## Step 7 – Histogram + KDE
```

In [10]:
```
plt.figure(figsize=(10, 4))
sns.histplot(data=df_241UDS0109, x="rentals", bins=15, kde=True)
plt.title("Bike Rental Distribution with Histogram + KDE")
plt.xlabel("Number of Rentals")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



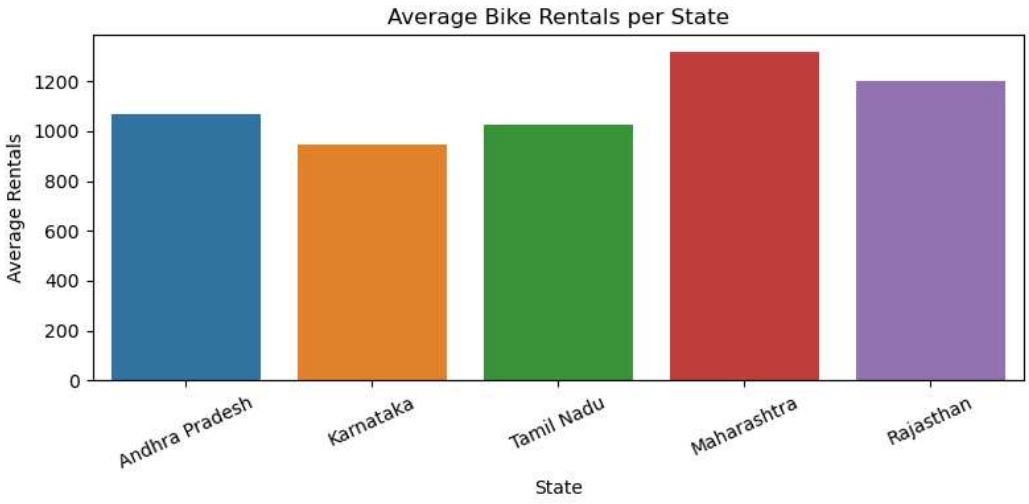
In [ ]:
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## Step 8 – Count Plot
### Number of Data Points for Each State
```

In [11]:
```
plt.figure(figsize=(8, 4))
sns.countplot(data=df_241UDS0109, x="state", hue="state", legend=False)
plt.title("Number of Data Points per State")
plt.xlabel("State")
plt.ylabel("Count")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```



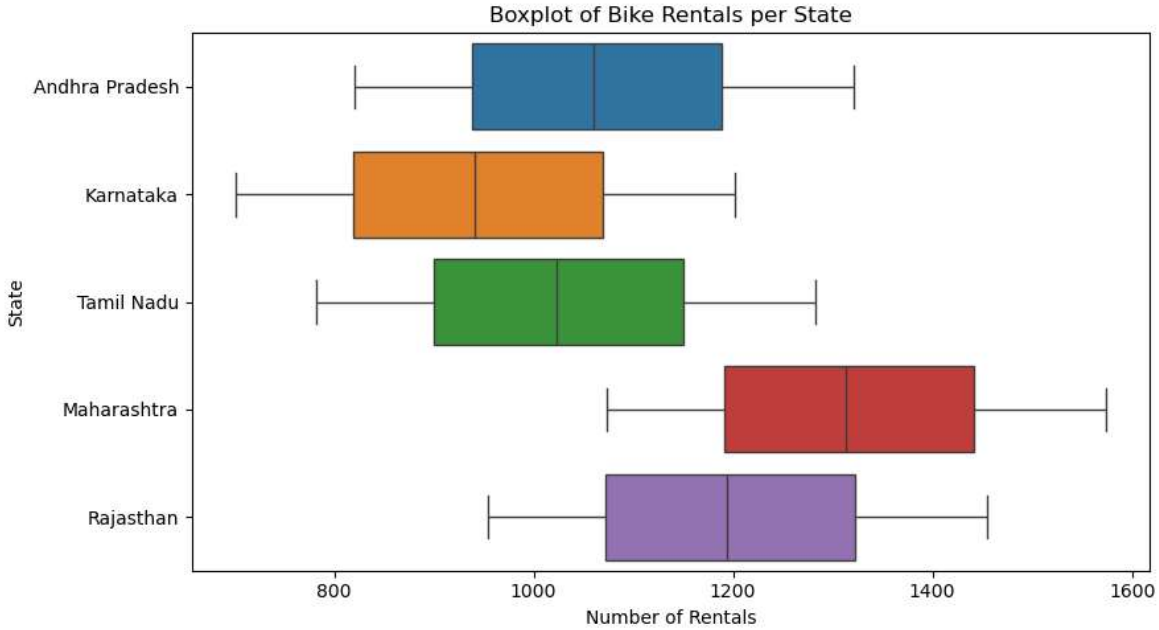
In [ ]:
```
## Step 9 – Bar Plot
### Average Bike Rentals per State
```

In [12]:
```
plt.figure(figsize=(8, 4))
sns.barplot(data=df_241UDS0109, x="state", y="rentals", hue="state", estimator=np.mean, errorbar=None,
legend=False)
plt.title("Average Bike Rentals per State")
plt.xlabel("State")
plt.ylabel("Average Rentals")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```



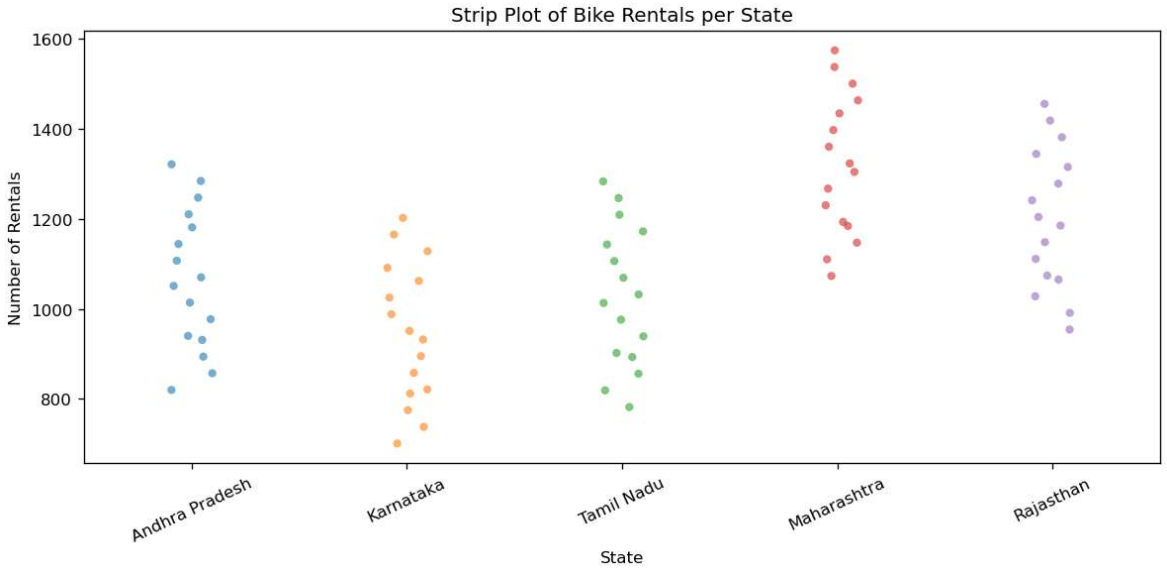
In [ ]:
```
## Step 10 – Boxplot
### Finding IQRs and Outliers of Rentals in Different States
```

In [13]:
```
plt.figure(figsize=(9, 5), dpi=100)
sns.boxplot(data=df_241UDS0109, x="rentals", y="state", hue="state", legend=False)
plt.title("Boxplot of Bike Rentals per State")
plt.xlabel("Number of Rentals")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```



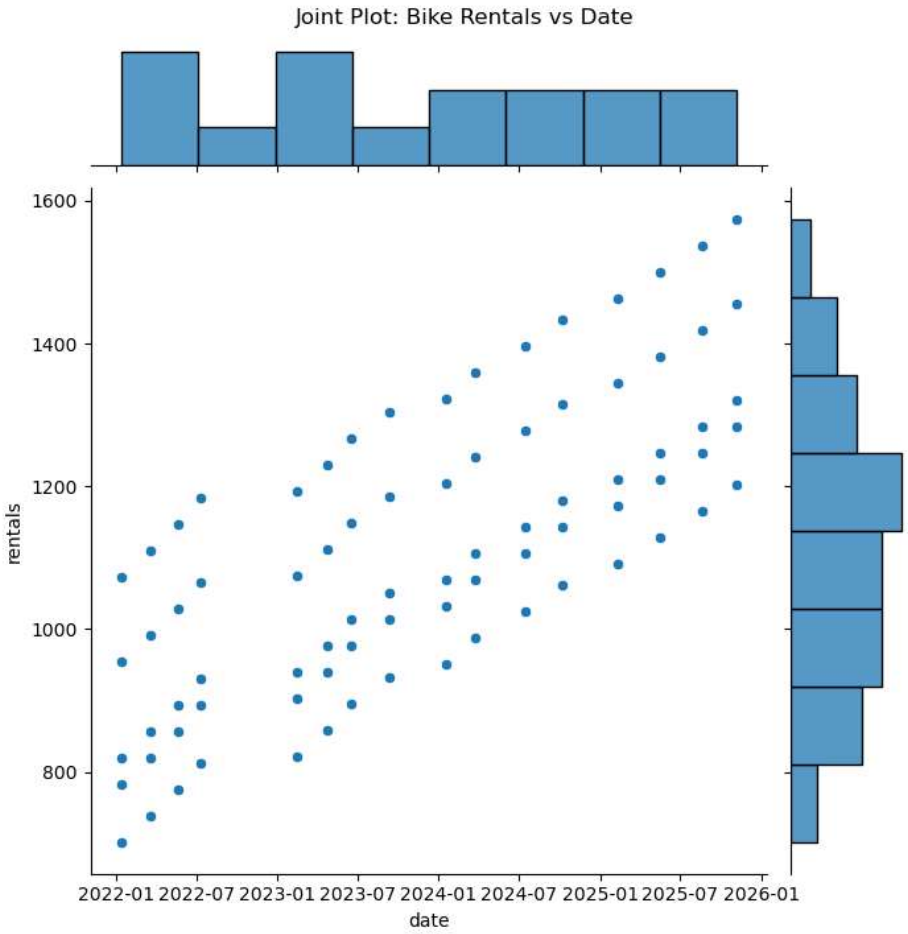
In [ ]:
```
## Step 11 – Swarm / Strip Plot
### Rental Distribution in Different States
```

In [14]:
```
plt.figure(figsize=(10, 5), dpi=120)
sns.stripplot(data=df_241UDS0109, x="state", y="rentals", hue="state", alpha=0.6, legend=False)
plt.title("Strip Plot of Bike Rentals per State")
plt.xlabel("State")
plt.ylabel("Number of Rentals")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```



In [ ]:
```
## Step 12 – Joint Plot
### Finding Distributions with Joint Plot
```

In [15]:
```
sns.jointplot(data=df_241UDS0109, x="date", y="rentals", height=7)
plt.suptitle("Joint Plot: Bike Rentals vs Date", y=1.02)
plt.show()
```



In [ ]:
```
## Step 13 – Heat Map Preparation
### Average Rentals per Year per State
```

In [16]:
```
yearly_avg_241UDS0109 = df_241UDS0109.groupby(
    ["year_241UDS0109", "state"]
)["rentals"].mean()
yearly_avg_241UDS0109
```

Out[16]:
```
year_241UDS0109  state
2022             Andhra Pradesh     875.5
                 Karnataka          756.5
                 Maharashtra       1128.5
                 Rajasthan         1009.5
                 Tamil Nadu         837.5
2023             Andhra Pradesh     995.5
                 Karnataka          876.5
                 Maharashtra       1248.5
                 Rajasthan         1129.5
                 Tamil Nadu         957.5
2024             Andhra Pradesh    1125.5
                 Karnataka         1006.5
                 Maharashtra       1378.5
                 Rajasthan         1259.5
                 Tamil Nadu        1087.5
2025             Andhra Pradesh    1265.5
                 Karnataka         1146.5
                 Maharashtra       1518.5
                 Rajasthan         1399.5
                 Tamil Nadu        1227.5
Name: rentals, dtype: float64
```

In [ ]:
```
## Step 14 – Using unstack()
Reshape the grouped result so each row represents a year and each column represents a state.
```

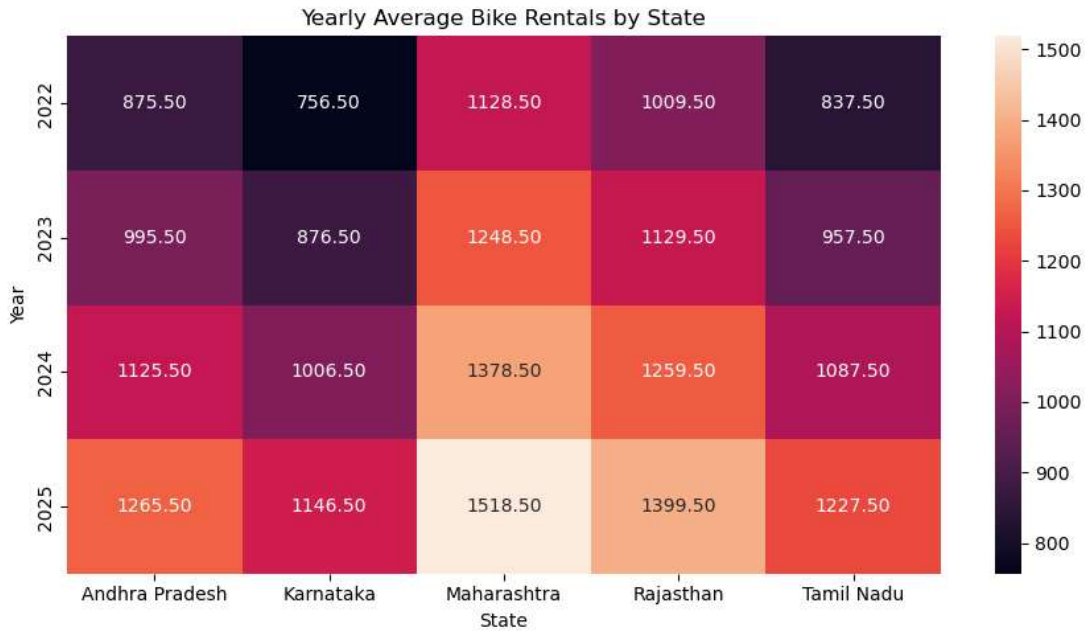
In [17]:
```
yearly_avg_241UDS0109 = df_241UDS0109.groupby(
    ["year_241UDS0109", "state"]
)["rentals"].mean().unstack()
yearly_avg_241UDS0109.head()
```

Out[17]:

state	Andhra Pradesh	Karnataka	Maharashtra	Rajasthan	Tamil Nadu
year_241UDS0109					
2022	875.5	756.5	1128.5	1009.5	837.5
2023	995.5	876.5	1248.5	1129.5	957.5
2024	1125.5	1006.5	1378.5	1259.5	1087.5
2025	1265.5	1146.5	1518.5	1399.5	1227.5

In [ ]:
```
## Step 15 – Heatmap
### Yearly Average Bike Rentals by State
```

In [18]:
```
plt.figure(figsize=(9, 5), dpi=100)
sns.heatmap(yearly_avg_241UDS0109, fmt=".2f", annot=True)
plt.title("Yearly Average Bike Rentals by State")
plt.xlabel("State")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```



In [ ]:
```
RESULT:-
Various Seaborn visualization techniques were applied to analyze and compare city bike rentals across states and
years using a new dataset.
```